

Alec Bossard

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Professional Summary

Final-year Robotics Engineering student with AIST research in 3D perception and Vision-Language Robotics. Builds AI and robot software using Python, PyTorch and ROS. Seeking a final-year internship from March to August 2027.

Education

UPSSITECH - Université de Toulouse

Engineering Degree in Robotics - Systèmes Robotiques et Interactifs | Sep 2024 - Jun 2027 (expected)

Experience

AIST Japan - AIRC Computer Vision Research Team

Robotics Research Intern - 3D Perception & Vision-Language Robotics | Tsukuba, Japan | Apr 2026 - Aug 2026

- Built an open-vocabulary RGB-D pipeline with Grounding DINO object detection and SAM2 segmentation.
- Extended SVLR with RGB-D and Vision-Language Model (VLM) perception for long-horizon manipulation.
- Contributed to 3D object localization and persistent world memory for robotic manipulation.

TLSe Racing - Formula Student Driverless Team

Software Developer - Driverless Team | Toulouse, France | Sep 2025 - Present

- Reduced simulated lap time from 44 s to 30 s using RRT* planning, spline smoothing, curvature-aware control, adaptive target speed and variable lookahead.
- Integrated the planning stack into ROS/Gazebo using vehicle odometry and cone positions to generate real-time commands for an autonomous race car.
- Developed local planning for track discovery and full-track trajectory generation for mapped circuits; compared planning and control strategies in simulation.

Projects

EarlyEval Reproduction

Independent reduced reproduction | Python, LightGBM, scikit-learn

- Built an agent-outcome prediction pipeline from partial execution traces; evaluated 700 trajectories across 350 shared SWE-bench Verified tasks.
- Trained two LightGBM models on behavioral and TF-IDF/SVD text features, then applied Platt calibration.
- Conducted held-out-agent evaluation and controlled ablations; documented reproducible findings, including negative results, with unit tests, integration tests and auditable experiment outputs.

Path Planning Studio

Autonomous racing prototyping platform | TypeScript, React, Three.js

- Built an interactive 3D platform with RRT* planning, trajectory optimization, vehicle simulation and live telemetry for prototyping before ROS2 integration.
- Reduced offline analysis from about 45 to 20 minutes per setup iteration using interactive racing-line visualization.

Technical Skills

Programming: Python, C++, TypeScript, JavaScript

Robotics: ROS, ROS2, Gazebo, path planning, RRT*, trajectory generation, vehicle control, robotic manipulation

AI & Computer Vision: PyTorch, scikit-learn, LightGBM, OpenCV, RGB-D perception, object detection, segmentation, Vision-Language Models

Software & Research Tools: React, Three.js, NumPy, SciPy, Git, Linux, pytest, Ruff